

Here are your Kerberos notes in a clean, professional format. You can copy and paste this directly into Google Docs or Microsoft Word.

Kerberos – Short Notes

Definition

Kerberos is a network authentication protocol. It provides secure authentication between users and services using a trusted third party called the **KDC (Key Distribution Center)**.

Key Features

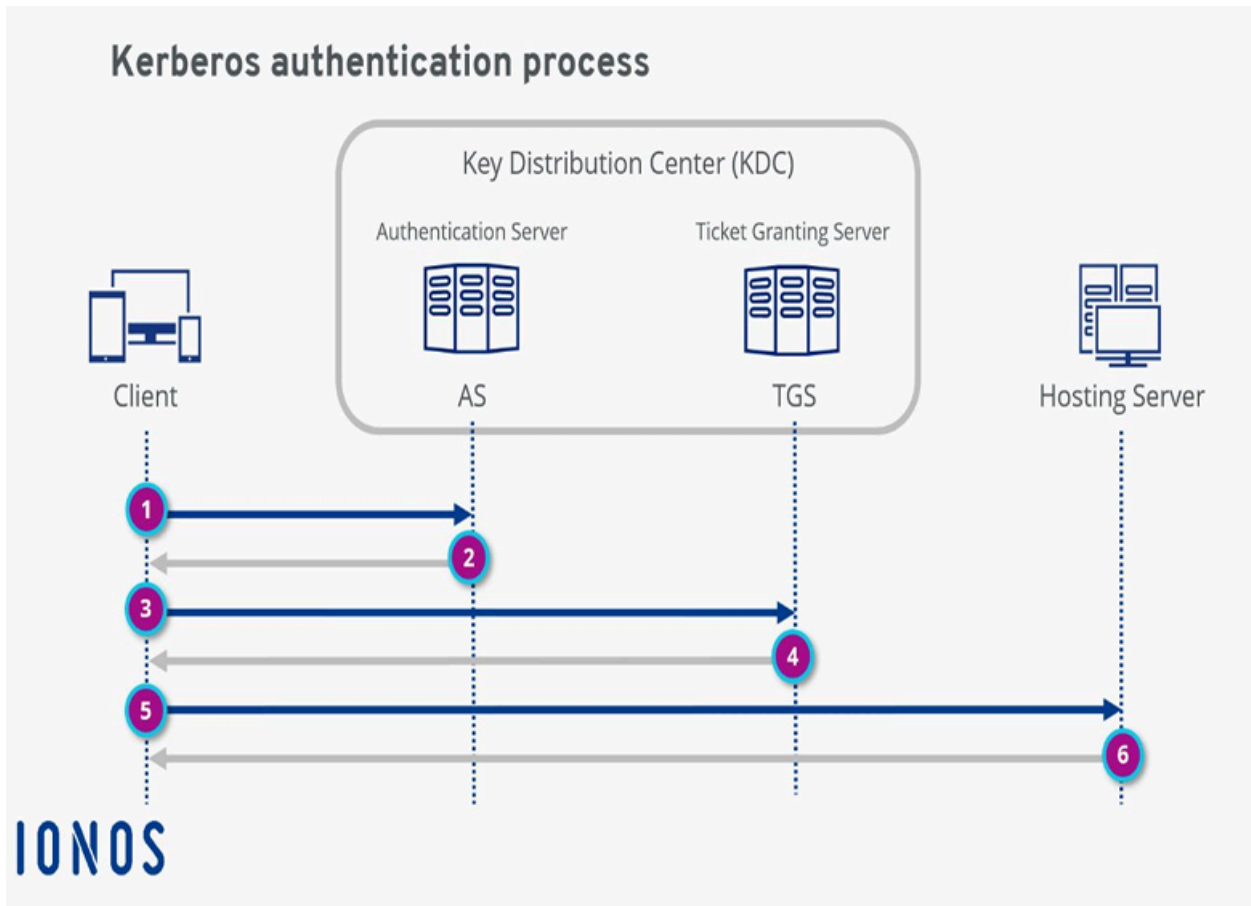
- **Symmetric Key Cryptography:** Uses shared secret keys for encryption.
- **Password Security:** The user's password is never sent over the network.
- **Secure Environment:** Provides a highly secure login system for untrusted networks.

Main Components

- **KDC (Key Distribution Center):** The central trusted authority responsible for authentication.
- **Authentication Server (AS):** Verifies the user's identity and issues the Ticket Granting Ticket (TGT).
- **Ticket Granting Server (TGS):** Processes requests for specific service access using the TGT.
- **Ticket Granting Ticket (TGT):** A temporary credential used to request access to various services without re-entering credentials.
- **Service Ticket:** A specific ticket used to access a particular service or server.
- **Session Key:** A temporary cryptographic key used for secure communication during a session.

Working (Steps)

1. **Request:** The Client sends an authentication request to the **AS**.
2. **Verify & TGT:** The AS verifies the user and sends back a **TGT** and a **session key**.
3. **TGS Request:** The Client sends the TGT to the **TGS** to request a specific service.
4. **Service Ticket:** The TGS verifies the TGT and provides a **Service Ticket**.
5. **Service Request:** The Client sends the Service Ticket to the target **Server**.
6. **Access:** The Server verifies the ticket and grants the client access.



Advantages

- **Strong Security:** Utilizes high-level encryption and timestamps to prevent replay attacks.
- **Single Sign-On (SSO):** Users only need to log in once to access multiple services.
- **No Password Transmission:** Eliminates the risk of password interception since they are never sent over the wire.

One-Line Summary (for revision)

Kerberos is a secure authentication system using tickets and a trusted KDC to verify users without sending passwords over the network.
